

PROFESSIONAL SUMMARY

Electronics and computer engineer with strong software engineering skills. Builds production-grade web apps, embedded systems, and AI tools. Experienced with React and Node, machine learning for trading tools, and real-time systems. Comfortable across the stack from PCB and sensors to cloud and Docker. Seeking a junior to middle engineer role where I can deliver measurable impact quickly.

CORE SKILLS

Programming: Python, JavaScript, TypeScript, Java, C++, C#, C, Rust

Frameworks and Tools: React, Node, TensorFlow, Docker, Google Cloud, GitHub Actions, Pine Script, ROS, OpenCV, Arduino, Supabase

Data and AI: KNN, Perceptron, Gaussian filtering, NEAT, signal processing, computer vision

Embedded and Hardware: LoRa, RF, BLE, PWM, PID control, sensors, TENS, EEG, PCB design, 3D printing

Web and Mobile: React, Ionic, Capacitor, p5.js, TypeScript, SQL, SQLite, Bun, TailwindCSS, Electron, Vite, Shadcn UI

Networking: TCP/IP, routing, switching, subnetting, Cisco device configuration, network security

Other: Stripe payments, subscriptions, MongoDB, real-time telemetry

EDUCATION

Politechnika Wroclawska

Oct 2021 – Feb 2026

Bachelor of Engineering in Electronics and Computer Engineering (GPA: 4.2)

- **Relevant Coursework:** Artificial Intelligence (Python, JS), Software Engineering (JS, Java, C++), Embedded Systems, Circuit and Signal Theory, Physics, Mobile & Web Development (JS, Java, C++).

Cisco Networking Academy

Oct 2023 – Feb 2024

CCNA: Introduction to Networks training for associate-level jobs

- **Relevant Coursework:** TCP/IP and Network Protocols, IP Addressing and Subnetting, Routing and Switching, Network Security Essentials, Cisco Device Configuration

EXPERIENCE

TradesCove (Wroclaw, PL)

Nov 2025 – Current

Founder / Software Engineer

- Developed and launched a subscription-based website offering an AI-powered trading indicator using React.js & Node.js.
- Designed and implemented ML models (KNN, Perceptron) with custom distance functions to generate trading signals.
- Created an advanced, high-performance charting library for interactive market visualization.
- Architected and fully managed the PostgreSQL database (schema design, indexing, migrations, performance tuning).
- Built a full internal admin panel to manage end-to-end operations: schedules, organizations, users, emails/broadcasts, database tools, and marketing content.
- Served as CEO/founder, owning end-to-end execution across product, engineering, operations, finance, customer support, and marketing/growth.

Blebox (Wroclaw, PL)

Oct 2025 – Nov 2025

Software Engineer, Embedded Engineer Intern

- Built a Rust backend that ingests and serves real-time and day-ahead electricity prices via reliable APIs for low-latency decisioning.
- Developed ESP32 firmware to fetch price signals and compute control/optimization values tailored to the connected grid/PV electrical system.
- Created an Electron desktop app to manage device connectivity, visualize usage and costs, and surface insights in a unified dashboard.
- Implemented scheduling/optimization logic that recommends the cheapest time windows to use electricity and supports day-to-day operations.

Software Engineer

- Developed and launched a subscription-based website offering an AI-powered trading indicator using React.js & Node.js.
- Designed and implemented AI models (KNN, Perceptron). With custom distance functions to power the trading indicator.
- Optimizing MongoDB database and Stripe customers reduced query response time by 60%.
- Integrated Stripe for secure payment processing and managed user subscriptions.
- Managed business operations, customer support, product iterations, and marketing.

Istanbul Technical University (Istanbul, Turkey)

Dec 2019 – Jul 2020

Research Assistant

- Designed and implemented embedded control systems for a high-altitude data collection platform using C++.
- Developed real-time telemetry and data logging software for environmental sensors.
- Integrated wireless communication modules (LoRa, RF) for long-range data transmission.
- Created ground station software for live monitoring and post-flight analysis using Python and custom GUIs.
- Optimized power management systems to extend operational life in harsh environmental conditions.

PROJECTS

Artificial Intelligence Trading Tool

Technologies: React.js, Pine Script, JS, Machine Learning

- Developed an AI-based trading indicator designed to predict market trends and assist traders in making data-driven decisions.
- Implemented Machine Learning algorithms, including KNN, Perceptron, and Custom Gaussian filtering, to analyze historical market data and generate real-time predictions.
- Improved prediction accuracy of the KNN algorithm by 70% using custom-fitted distance functions.

N.E.A.T. AI Web Game

Technologies: JS, p5.js, N.E.A.T.

- Developed a web game where players control a spaceship to avoid asteroids, with the game fully running on AI-powered agents.
- Used N.E.A.T. to train AI agents that autonomously control the spaceship.
- Designed the game's graphics and animations using p5.js.
- Focused on improving agent performance with custom fitness functions that evolve over time.

Autonomous Mars Rover

Technologies: Python, ROS, OpenCV, Arduino, 3D

- Built a multi-terrain rover capable of autonomous navigation using real-time AI simulation.
- Designed a soil collection system using a robot arm to gather samples for chemical analysis.
- Implemented a protein detection system using copper-based reagents to identify potential signs of life.
- Integrated computer vision and sensor fusion for terrain analysis and obstacle avoidance.

High-Altitude Data Collection Vehicle

Technologies: C++, Arduino, Sensors, Radio Comms, 3D

- Developed a single-EDC flying vehicle designed to ascend to high altitudes and gather environmental data.
- Implemented a dual-motor system spinning in opposite directions to counteract gyroscopic yaw for stable flight.
- Integrated multiple sensors and shielding for interference from high-current BLDC motors.
- Real-time data transmission using high-power RF transmitters.
- Increased data transmission stability by 40% and decreased noise by 75% by integrating a custom object-tracking yagi antenna.

Brainwave-Controlled Muscle Stimulation

Technologies: EEG, PCB, Signal Processing, TENS, Embedded Systems

- Designed and built a custom EEG device to read brainwaves using a custom board with multiple amplifiers and signal filtering.
- Built a system that translates brain activity into signals to control another person's muscles via a TENS unit.
- Engineered real-time signal processing to extract brainwave patterns for activation of preselected muscles.

Mootify - Emotion Sharing Watch

Technologies: Ionic Capacitor, Supabase, BLE, Embedded Systems, Computer Graphics

- Created an Android/iOS app allowing users to send emotions to friends via a custom smartwatch.
- Implemented real-time notifications, storing interactions in a Supabase database.
- Developed BLE communication for the longevity of the Mootify watch battery and available screen time.

Self-Balancing Robot

Technologies: C, PWM, PID Control, Bluetooth, Ionic React Capacitor

- Engineered a fully autonomous balancing robot using a gyroscope-driven PID controller to maintain stability on two wheels.
- Implemented a real-time motor control system with PWM adjustments and direct register manipulation for optimized performance.
- Built a custom Bluetooth communication protocol, enabling control via a mobile app developed with Ionic React Capacitor.
- Enhanced system responsiveness by fine-tuning PID parameters and optimizing sensor data processing for real-time adjustments.

SKILLS

- **Languages:** English, Turkish, Japanese, Polish
- **Programming Languages:** Python, JavaScript, Java, C++, C#, Rust
- **Frameworks and Tools:** React.js, Node.js, TensorFlow, Docker, Google Cloud, GitHub Actions, Rust
- **Interests:** Machine Learning, Unity 3D, Unreal Engine, Game Development, 3D design, Blender 3D, MGFA, Piano